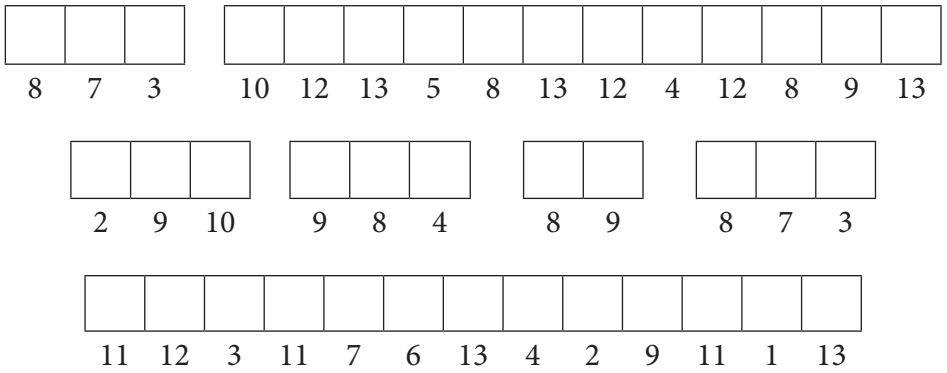
WORKSHEET

Chapter 3: Transformations of absolute values and hyperbolas

I've learned from experience that the greater part of our happiness or misery depends on …



Unknown

Questions

Describe how the following transformations affect the graphs of $|x|$ and $\frac{1}{x}$.

Example: For $|x + 3| - 5$, the graph is moved 3 units to the left and 5 units down.

1 $\frac{1}{x} - 3$

2 $|x + 5|$

3 $\frac{1}{x - 2} + 1$

4 $|x + 2| + 1$

5 $\frac{1}{x - 4} + 5$

6 $|-3 + x| - 2$

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7 $\dfrac{1}{4+x} - 5$

8 $|x-2|$

9 $\dfrac{1}{x} + 5$

10 $|x+1| - 3$

11 $\dfrac{1}{x-4} - 6$

12 $|x+3| + 2$

13 $\dfrac{1}{x+5} - 1$

Solutions

A 5 units left

C 4 units right, 6 units down

D 1 unit left, 3 units down

E 3 units down

I 3 units left, 2 units up

M 3 units right, 2 units down

N 5 units up

O 2 units right

P 4 units right, 5 units up

R 2 units right, 1 unit up

S 5 units left, 1 unit up

T 2 units left, 1 unit up

U 4 units left, 5 units down

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